time → energy → macroscopic state[81]

That is: to define the macroscopic state, we first need to know the energy, and to define energy we first need to know what is time. In this logic, time comes first and is independent from the rest.

But there is another way of thinking about this same relationship: by reading it in reverse. That is, to observe that a macroscopic state, which is to say a blurred vision of the world, may be interpreted as a mingling that preserves an energy, and this in its turn generates a time. That is:

macroscopic state → energy → time[82]

This observation opens up a new perspective: in an elementary physical system without any privileged variable that acts like "time"—where, in effect, all the variables are on the same level but we can have only a blurred vision of them described by macroscopic states. A generic macroscopic state *determines* a time.

I'll repeat this point, because it is a key one: a macroscopic state (which ignores the details) chooses a particular variable that has some of the characteristics of time.

In other words, a time becomes determined simply as an effect of blurring. Boltzmann understood that the behavior of heat involves blurring, from the fact that inside a glass of water there is a myriad of microscopic variables that we do not see. The *number* of possible microscopic configurations for water is its entropy. But something further is also true: the blurring itself determines a particular variable—time.

In fundamental relativistic physics, where no variable plays a priori the role of time, we can reverse the relation between macroscopic state and evolution of time: it is not the evolution of time that determines the state, it is the state—the blurring—that determines a time.

Time that is determined in this way by a macroscopic state is called "thermal time." In what sense may it be said to be a time? From a microscopic point of view, there is nothing special about it—it is a variable like any other. But from a macroscopic one, it has a crucial characteristic: among so many variables all at the same level, thermal time is the one with behavior that most closely resembles the variable we usually call "time," because its relations with the macroscopic states are exactly those that we know from thermodynamics.

But it is not a universal time. It is determined by a macroscopic state, that is, by a blurring, by the incompleteness of a description. In the next chapter, I